

Finssentials - Architecture Overview

1. What is Finssentials?

Finssentials is a **web-based financial analysis platform** for M&A and Financial Due Diligence:

- **Reporting** on general-ledger data (P&L, balance sheet, cash flow, working capital, sales)
- **Data ingestion** (Excel/GDPdU upload, mapping, versioning)
- **FDD Bot** — guided workflow with upload, configuration, and Excel outputs
- Optional: additional **chat assistants** (Expert, Exit Readiness)

Data: Financial data and client uploads (Excel) — **confidential, GDPR-relevant**. In a later stage also **ERP data** (direct integration).

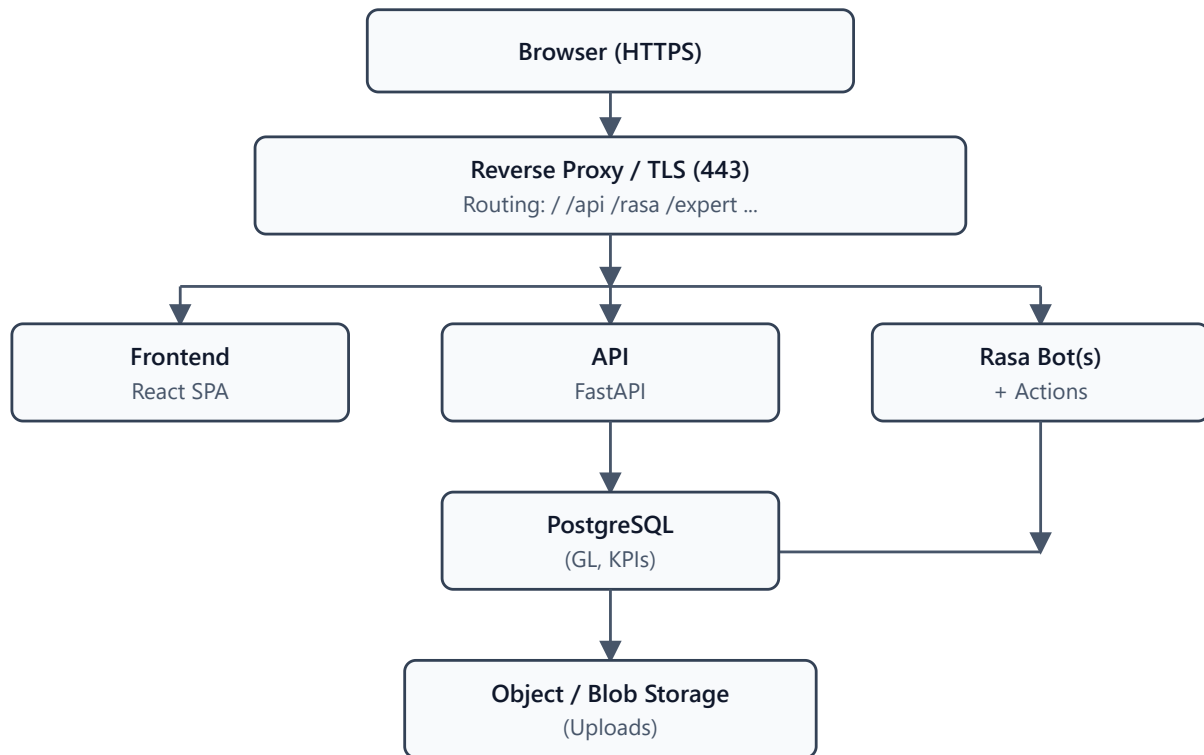
Audience: Advisory teams and their clients (B2B).

2. Product variants (please clarify in the call which variant is planned)

	Variant A — Full stack	Variant B — Lean (GDPdU / FDD)
Scope	Cockpit, Financials, Sales, Exit Readiness, FDD Bot	Financials, Data Update, Mapping, FDD Bot
Chatbots	3× Rasa (FDD + Expert + Readiness)	1× Rasa (FDD)
Containers (approx.)	9 services + reverse proxy	5 services + reverse proxy
Typical entry	Production / larger teams	Pilot / lean rollout

Both variants share the **same technical foundation** (React, FastAPI, PostgreSQL, Docker).

3. Architecture (logical)



Optional: AI Gateway -> Enterprise LLM (Azure OpenAI / Bedrock), EU region

4. Component checklist

Please quote **staging** + **production** separately (2× environments).

#	Component	Requirement	Sizing note
1	PostgreSQL	Managed, EU (DE/FI/SE), backups, PITR	Core DB; start ~64–256 GB storage
2	Application hosting	Docker Compose or equivalent	Full stack: 8 vCPU / 32 GB RAM ; Lean: 4 vCPU / 16 GB
3	Frontend	Static SPA (React build)	Same host or CDN
4	Reverse proxy / TLS	HTTPS, routing to API + bots	e.g. Caddy, NGINX, App Gateway
5	Object storage	Uploads & Excel outputs	Start ~100–500 GB per environment
6	Identity (SSO)	OIDC / Entra ID, MFA	For production
7	Secrets	Vault or similar	DB passwords, API keys
8	Monitoring & logs	Health checks, alerts, 30–90 day retention	/health , Rasa /status
9	Backups & restore	DB + optional blob; restore test	GDPR / ISO readiness
10	Network	DB not public; private endpoints	Firewall / VPN for admin
11	CI/CD integration	Pull from GitHub Container Registry	No manual FTP deploy
12	Optional: AI	LLM proxy + enterprise endpoint	No raw client data to public APIs

5. Rough scale (initial — not maximum)

Parameter	Planning assumption
Internal users	10–20
Clients / engagements	5–15 (growth expected)
Concurrent users	5–10
FDD sessions / month	~200–500
Upload size	typically 5–50 MB per file
Availability	99.5%+ (business hours initially OK)

6. Compliance & location

Topic	Requirement
Data residency	EU (preferably DE, alternatively FI/SE/FR)
DPA	Required
GDPR	Client data, deletion & retention concept
ISO 27001	Path to certification (not required day one)
Sub-processors	List + transparency

7. Five-year scalability plan (Financial Model)

7.1 Active customer base (forecast)

Year (Dec)	Total	FDD Bot subscription	Reporting tool	Exit Readiness	All-in-One
2026	9	3	6	0	0
2027	61	18	18	12	13
2028	132	39	33	30	31
2029	235	71	51	56	57
2030	388	120	73	97	98

7.2 Infrastructure scaling requirements (derived)

Year (Dec)	Active customers	Tenants with GL data ¹	Peak concurrent users ²	FDD sessions / month ³	PostgreSQL (prod) ⁴	Object storage (prod) ⁴	Application tier ⁵
2026	9	6	6	~15	64 GB	~200 GB	4 vCPU / 16 GB
2027	61	43	40	~125	200 GB	~1.2 TB	8 vCPU / 32 GB
2028	132	93	80	~280	400 GB	~2.5 TB	8 vCPU / 32 GB (+ read replica optional)
2029	235	164	140	~510	700 GB	~4.5 TB	16 vCPU / 64 GB or 2× app nodes
2030	388	268	235	~870	1–1.5 TB	~7 TB	2× app nodes (8 vCPU / 32 GB each) + DB replica

Planning assumptions:

1. **Tenants with GL data** = Reporting + Exit Readiness + All-in-One (products that store and derive general-ledger data).
2. **Peak concurrent users** $\approx$ 4 named users per customer $\times$ 15 % concurrent peak (advisory + client users).
3. **FDD sessions / month** $\approx$ 4 runs per month per FDD Bot subscription and All-in-One customer.
4. **Storage** = PostgreSQL incl. derived KPI tables & snapshots; object storage = uploads + Excel outputs (~8 GB/customer/year + ~15 GB/GL tenant). Rounded; quote with growth headroom (+30 %).
5. **Application tier** = combined API + Rasa + actions host(s); frontend static/CDN separate.